

Learning Activity Sheet for General Mathematics

Quarter 3

Unit

7

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Learning Activity Sheet for General Mathematics
Quarter 3: Lesson 7

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	7.1	Date:	
Lesson Title/ Topic:	Trigonometric Ratios in a Right-Angled Triangle Using a Real-Life Object or Figure		
Name:		Grade & Section:	

I. Activity No. 1: Practicing Your Skills in Trigonometric Ratios

II. Objectives:

At the end of the activity, you should be able to:

1. Find the unknown sides or angles in a right triangle.
2. Use Trigonometric ratios to solve problems involving a real-life object or figure.

III. Materials Needed:

pen and paper

IV. Instructions:

Perform the indicated tasks. Write your answer on a sheet of paper.

A. Use your calculator to find the value of the following, correct to two decimal places.

- | | | |
|--------------------|--------------------|--------------------|
| 1. $\cos 85^\circ$ | 3. $\cos 65^\circ$ | 5. $\tan 63^\circ$ |
| 2. $\sin 7^\circ$ | 4. $\tan 35^\circ$ | |

B. Find the size of the angle θ (to the nearest degree) where θ is acute.

- | | |
|--------------------------|--------------------------|
| 1. $\sin \theta = 0.529$ | 4. $\cos \theta = 0.401$ |
| 2. $\cos \theta = 0.493$ | 5. $\tan \theta = 2.5$ |
| 3. $\tan \theta = 1.98$ | |

C. Solve the following problems.

1. Triangle BCA is right-angled at C if $c = 27$ and $\angle A = 58^\circ$, find $\angle B$, b , and a .
2. Solve the remaining sides and angles of the right triangle ACB and right angled at C.

a. $a = 5\text{cm}$	$c = 8\text{ cm}$
b. $A = 54^\circ$	$b = 3\text{ cm}$

V. Synthesis:

Answer the following questions:

1. What do you notice about the values of sine, cosine, and tangent as the angle increases?
2. How can the calculator help verify your manual trigonometric computations?
3. Why is it important to round answers consistently in trigonometric calculations?
4. How do trigonometric ratios help you find missing sides and angles in a right triangle?
5. When is it more convenient to use sine, cosine, or tangent?
6. What is the importance of correctly labeling the sides relative to the given angle?
7. How can this method be applied to real-life measurement problems (e.g., height of a tree or building)?

I. Activity No. 2: Estimating the Height Using Trigonometric Ratios

II Objectives:

At the end of the activity, you should be able to:

1. Find the unknown sides or angles in a right triangle.
2. Use trigonometric ratios to solve problems involving a real-life object or figure.

III. Materials Needed:

- Protractor (or smartphone with angle-measuring app)
- Measuring tape
- String or stick (to help sight the angle)
- Calculator

IV. Instructions:

1. Form a group of 5 members.
2. Prepare the materials:
3. Stand at a known distance (e.g., 10 meters) from the base of the tree or flagpole.
4. Use the protractor to measure the angle of elevation from your eye level to the top of the tree.
5. Record your eye-level height (e.g., 1.6 meters).
6. Use an appropriate trigonometric ratio, to find the height of the tree.

V. Synthesis: Answer the following questions:

1. Which trigonometric ratio did you use to solve this problem, and why?
2. How does the angle of elevation affect the height of the kite?
3. What real-life factors could make your calculation slightly different from the actual height?
4. In what other real-life situations can you use the same trigonometric principle?

I. Activity No. 3: Checking Understanding on Trigonometric Ratios

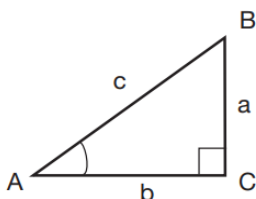
II. Objectives: At the end of the activity, you should be able to:

1. Find the unknown sides or angles in a right triangle.
2. Use trigonometric ratios to solve problems involving a real-life object or figure.

III. Materials Needed: pen and paper

IV. Instructions: Read and understand the following problems. Solve each item.

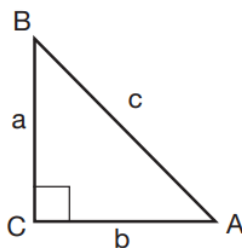
A. Using the figure below, write the expressions that give the required unknown value.



- | | |
|--|--|
| 1. If $A = 15^\circ$ and $c = 37$, find a . | 6. If $A = 19^\circ$ and $a = 11$, find c . |
| 2. If $A = 76^\circ$ and $a = 13$, find b . | 7. If $c = 16$ and $a = 7$, find b . |
| 3. If $A = 49^\circ$ and $a = 10$, find c . | 8. If $b = 10$ and $c = 20$, find a . |
| 4. If $a = 21.2$ and $A = 71^\circ$, find b . | 9. If $a = 7$ and $b = 12$, find A . |
| 5. If $a = 13$ and $B = 16^\circ$, find c . | 10. If $a = 8$ and $c = 12$, find B . |

B. Use the given figure to solve the remaining parts of right triangle ACB

1. $b = 17$ cm and $c = 23$ cm
2. $c = 16$ and $a = 7$
3. $b = 10$ and $c = 20$
4. $b = 6$ and $c = 13$
5. $c = 13$ and $a = 12$



C. Solve the following problems:

1. A kite string is 20 meters long and makes an angle of 40° with the ground. How high is the kite above the ground?
2. A ladder 8 meters long leans against a wall making an angle of 65° with the ground. How high up the wall does the ladder reach?
3. A wheelchair ramp is 5 meters long and rises 1.25 meters high. Find the angle of inclination of the ramp with the ground.

V. Synthesis: Answer the following questions:

1. Have you completed all the tasks correctly?
2. Which items did you find easy, and which did you find challenging?
3. What common errors did you make?
4. How will you address your misconceptions?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	7.2	Date:	
Lesson Title/ Topic:	AREA OF OBLIQUE TRIANGLES		
Name:		Grade & Section:	

I. Activity No. 1: Enhancing Skills in Finding the Area of Oblique Triangles

II. Objectives: At the end of the activity, you should be able to:

1. Solve the areas of oblique triangle using Heron's Formula and $A = \frac{1}{2}ab \sin C$.
2. Apply Heron's Formula and $A = \frac{1}{2}ab \sin C$ in real-life objects/contexts.

III. Materials Needed: pen and paper

IV. Instructions:

Solve the following problems individually. Show complete step-by-step solutions. Afterward, pair-check with your seatmate, then volunteer to explain one problem on the board.

Problem 1. A triangular fish pen in Laguna de Bay has sides 16 m, 18 m, and 25 m. Find its area.

Problem 2. During typhoon season, a barangay is constructing a triangular roof for an evacuation center. The roof beams measure $a = 20$ m, $b = 25$ m, with an included angle of 40° . Find the roof's triangular area.

Problem 3. A triangular rice field in Isabela has sides 40 m, 55 m, and 70 m. Find its area. Round to two decimal places.

Problem 4. In Cagayan Valley, engineers are reinforcing a flood control dike. A triangular concrete support has two sides measuring 32 m and 27 m, with an included angle of 55° . Find the area of the triangular support.

Problem 5. The barangay built a triangular stage with sides 10 m, 15 m, and 17 m. Find its area.

V. Synthesis: Answer the following questions:

1. Which problem was easiest or most challenging, and why?
2. How did working with your seatmate help you understand the steps better?
3. How did choosing the correct formula affect your solution?
4. How can the skills learned be applied in real-life situations like construction or agriculture?

I. Activity No. 2: Applying Triangles in Real-Life Design

II. Objectives: At the end of the activity, you should be able to:

1. Solve the areas of oblique triangle using Heron's Formula and $A = \frac{1}{2}ab \sin C$.
2. Apply Heron's Formula and $A = \frac{1}{2}ab \sin C$ in real-life objects/contexts.

III. Materials Needed: pen and paper

IV. Instructions:

Analyze the scenario and perform the indicated tasks.

Scenario:

Your barangay is planning community improvement projects: a small plaza, a new evacuation center, and optimizing local rice fields. Many of these projects involve triangular spaces: triangular garden plots in the plaza, triangular roof trusses for typhoon-resistant buildings, and irregular triangular-shaped rice fields.

Your group will act as community planners, engineers, or agricultural advisors to model these triangles, calculate their areas, and make practical decisions, including estimating the amount of materials needed and the cost involved.

- Plaza Garden Plot:
 - Each triangular garden plot needs soil, grass, or paving stones.
 - Assume 1 bag of soil covers 2 m^2 and costs ₱350 per bag.
 - Calculate how many bags are needed for your triangle and the total cost.
- Evacuation Center Roof Truss:
 - Each triangular truss covers a portion of the roof.
 - Assume 1 sheet of roofing material covers 2 m^2 and costs ₱500 per sheet.
 - Calculate how many sheets are needed for your truss and the total cost.
 - Discuss how the triangle's shape affects stability during typhoons.
- Rice Field Plot:
 - Farmers need fertilizer for their triangular rice field.
 - Assume 1 kg of fertilizer covers 10 m^2 and costs ₱50 per kg.
 - Calculate how much fertilizer is needed and the total cost for your triangle.

Your task is to measure, calculate, and interpret these triangles to make real-life recommendations for your barangay.

Procedures:

1. Form Your Group

- Work in a group of 3–5 members.
- Your teacher will provide a triangle data card (two sides and the included angle) representing a plaza plot, roof truss, or rice field.

2. Model the Triangle

- **Option A:** Draw the triangle on the ground using chalk, masking tape, or measuring tape.
- **Option B:** Use a simulation tool like GeoGebra, Desmos, or a map app to model the triangle.
- **Label all sides and angles clearly.**

3. Calculate the Area

- Use the appropriate formula.
- Record your calculation on the worksheet.

4. Estimate Materials and Cost

- Use the scenario's material coverage and costs to compute:
 - Plaza: Number of soil bags and total cost.
 - Roof: Number of roofing sheets and total cost.
 - Rice Field: Kilograms of fertilizer and total cost.

5. Present Your Work

- Share your triangle model, area calculation, material estimates, and costs with the class.
- Explain how your calculations help in planning, construction, or agriculture.

V. Synthesis: Answer these questions:

1. What did you learn about applying triangle geometry in real-life situations?
2. How can these calculations help in budgeting, planning, or resource allocation?
3. Why is knowing the area of triangular spaces important for efficiency, safety, and cost management?

I. Activity No. 3: Applying Triangles in Real-Life Design

II. Objectives: At the end of the activity, you should be able to:

1. Solve the areas of oblique triangle using Heron's Formula and $A = \frac{1}{2}ab \sin C$.
2. Apply Heron's Formula and $A = \frac{1}{2}ab \sin C$ in real-life objects/contexts.

III. Materials Needed: pen and paper

IV. Instructions:

Part I – Concept Recall

1. Define an oblique triangle and give one real-life example in your community.
2. Explain briefly when to use Heron's Formula and when to use the trigonometric rule $Area = \frac{1}{2}absinC$.
3. Complete the sentence: "We use Heron's Formula when _____, and we use $Area = \frac{1}{2}absinC$ when _____."

Part II – Real-Life Applications

Solve the following problems completely. Show step-by-step solutions and include units.

1. The DPWH is surveying a triangular lot needed for a road widening project in Quezon City. The lot has sides measuring 35 m, 40 m, and 50 m. Find its area so they know how much land to acquire.
2. After a powerful typhoon in Albay, a family is rebuilding their home with stronger triangular roof supports to withstand future storms. Two support beams measure 18 meters and 22 meters, forming an included angle of 42° . Using trigonometry, calculate the area of the triangular roof support.
3. The city government of Baguio is designing a small triangular park with sides 20 m, 26 m, and 30 m. Find the park's area so they can estimate grass and landscaping costs.
4. A company in Metro Manila is installing a triangular solar-powered billboard. Two sides measure 14 m and 19 m, with an included angle of 37° . Find its area. If each square meter costs ₱1,200 to install, how much is the total cost?

V. Synthesis: Answer the following questions:

1. A student solved for the area of a triangle with sides 20 m, 25 m, and 30 m using $Area = \frac{1}{2}absinC$. Is the formula used correct? Why? If it's incorrect, what formula should have been used?
2. If you only know two sides of a triangle and a non-included angle, can you directly apply Heron's Formula or $Area = \frac{1}{2}absinC$? Justify your answer.

LEARNING ACTIVITY SHEETS

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	7.3	Date:	
Lesson Title/ Topic:	Shapes Beyond the Surface: Application of Right and Oblique Triangles		
Name:		Grade & Section:	

I. Activity No. 1: Delving Deeper on Triangles

II. Objectives: At the end of the activity, you should be able to:

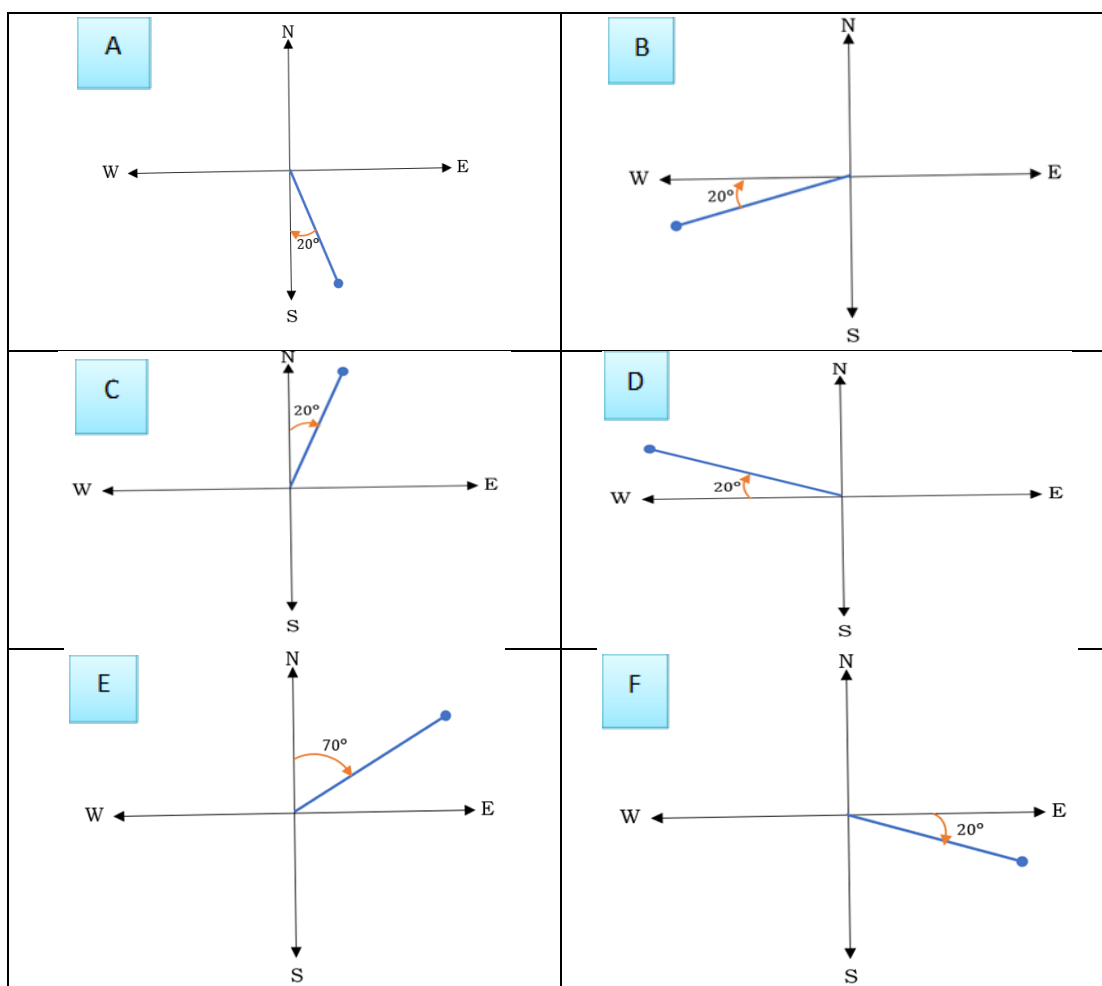
1. Complete the missing sides or angles of an oblique triangle
2. Apply the Laws of Sines and Cosines in solving problems on oblique triangles and in bearings in navigation.

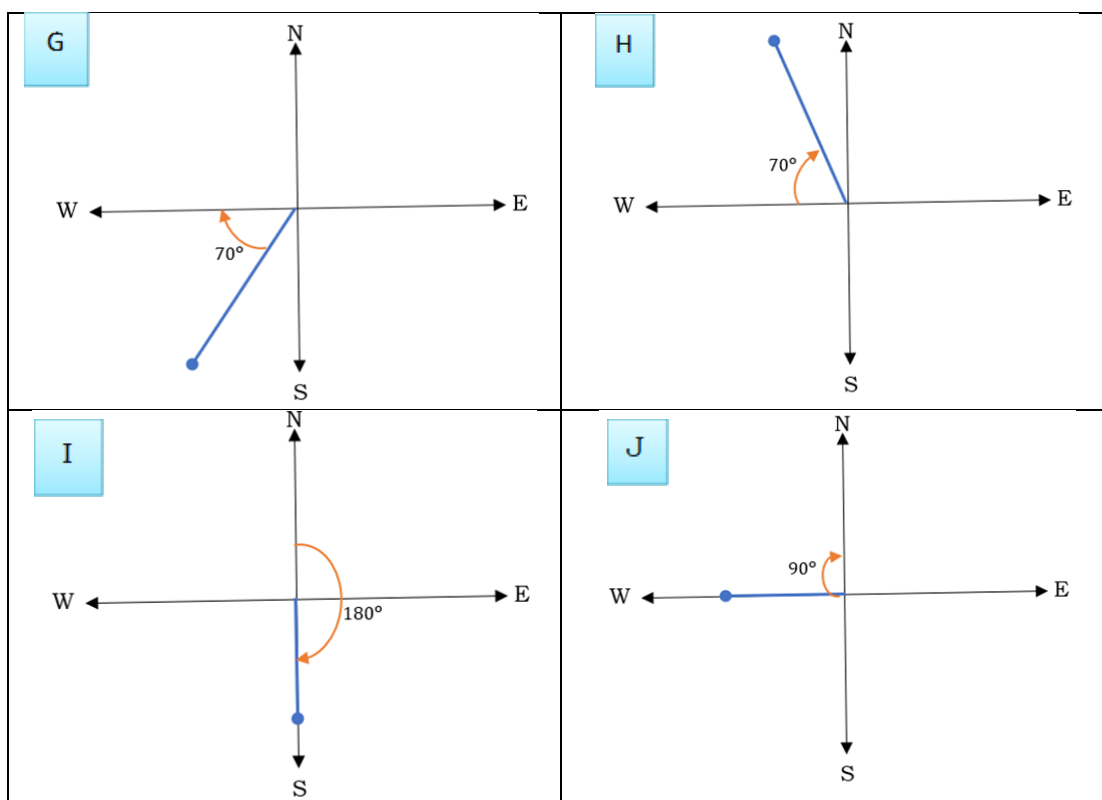
III. Materials Needed: pen and paper

IV. Instructions: Perform the indicated task.

Exercise 1. Match the measure of the given bearings with the appropriate graph below. Write the letter of your answer on the space before each number.

- | | |
|---------------------------|-------------------------|
| _____ 1. 20° | _____ 6. 340° |
| _____ 2. $S\ 70^\circ\ W$ | _____ 7. 110° |
| _____ 3. 160° | _____ 8. 270° |
| _____ 4. $S\ 20^\circ\ W$ | _____ 9. 180° |
| _____ 5. $N\ 70^\circ\ W$ | _____ 10. $N\ 70^\circ$ |





Exercise 2.

Solve the problems below.

- The legs of an isosceles right triangle are 1 unit long. Find the length of the longest side in simplest radical form.
- Use part (a) to find the exact value of each of the following. a. $\tan 45^\circ$ b. $\sin 45^\circ$ c. $\cos 45^\circ$
- An airplane is at an elevation of 35,000 ft when it begins its approach to an airport, its angle of descent is 6° . What is the distance between the airport and the point on the ground directly below the ground?
- A student looks out of a second-story school window and sees the top of the school flagpole at an angle of elevation of 22° . The student is 18 ft above the ground and 50 ft from the flagpole. Find the height of the flagpole.
- Observers at points A and B , 30 km apart, saw an airplane at angles of elevation of 40° and 75° , respectively. How far is the plane from each observer?
- A ship proceeds on a course of 300° for 2 hours at a speed of 15 knots (1 knot = 1 nautical mile per hour). Then it changes course to 230° , continuing at 15 knots for 3 hours. At that time, how far is the ship from its starting point?

Synthesis:

Answer the following questions:

- Which types of problems in this activity did you find easiest or most challenging, and why?
- How did understanding triangle properties and trigonometric ratios help you solve real-life navigation and distance problems?
- How can the skills and concepts practiced in this activity be applied in everyday situations, such as travel, construction, or aviation?

I. Activity No. 2: Mission Navigation: Find Your Way!

II. Objectives: At the end of the activity, you should be able to:

- Complete the missing sides or angles of an oblique triangle

2. Apply the Laws of Sines and Cosines in solving problems on oblique triangle and in bearings in navigation.

III. Materials Needed:

- Compass rose diagrams
- Graph paper or coordinate grids
- Protractors and rulers
- Calculators
- Scenario cards with bearing-based navigation tasks

IV. Instructions: Analyze the scenario below and perform the indicated task.

Scenario:

You are part of a rescue team tasked with locating a missing vessel based on bearing instructions. You must calculate positions, distances, and bearings using trigonometry and vector analysis.

Task:

A rescue boat leaves port on a bearing of 45° and travels 100 km. It then receives a signal to change course to 120° and travels another 80 km.

Goal: Determine the boat's final position and its bearing from the port.

Learners Roles:

Navigator: Calculates bearings and distances

Cartographer: Draws the path on a grid

Reporter: Explains the solution and reasoning

V. Synthesis: Answer the following questions:

1. What strategies did your team use to calculate bearings and distances, and how effective were they?
2. How did collaborating in different roles (Navigator, Cartographer, Reporter) help in solving the navigation problem?
3. How can the mathematical concepts you applied in this activity be used in real-life navigation or other practical situations?

I. Activity No. 3: Mission Navigation: Find Your Way!

II. Objectives: At the end of the activity, you should be able to:

1. Complete the missing sides or angles of an oblique triangle
2. Apply the Laws of Sines and Cosines in solving problems on oblique triangles and in bearings in navigation.

III. Materials Needed: pen and paper

IV. Instructions: Read and solve each problem

1. Find the measure of the largest angle in a triangle with sides having lengths $3\sqrt{6}$, $6\sqrt{3}$, and $9\sqrt{2}$.
2. Find the measure of the smallest angle in a triangle with sides 7, 12 and 13.
3. In $\triangle RST$, $\angle R = 75^\circ$, $\angle S = 45^\circ$ and $t = 3$. Find r and s .
4. In $\triangle ABC$, $\cos A = 0.6$. Find $\sin A$ and $\tan A$.
5. In $\triangle XYZ$, $\angle X = 90^\circ$, $\angle Y = 37^\circ$ and $z = 25$. Find x and y .
6. From points A and B, 10 meters apart, the angles of elevation of the top of the tower are 40° and 60° , respectively. Find the tower's height.

7. A ship leaves a port and sails northwest for 1 h and then northeast for 2 h. If it does not change speed, find what course the ship should take to return directly to port. Also find how long this return will take.

V. Synthesis: Answer the following questions:

1. Which types of problems in this activity did you find easiest or most challenging, and why?
2. How did understanding triangle properties and trigonometric ratios help you solve real-life navigation and distance problems?
3. How can the skills and concepts practiced in this activity be applied in everyday situations, such as travel, construction, or aviation?

Unit 7- Performance Task

Barangay Tri-Math-lon: Practical Applications of Trigonometry

Goal:

Use trigonometry and triangle-based calculations to plan community projects and solve navigation problems, demonstrating practical problem-solving skills.

Role:

You are a member of a barangay development and rescue team, acting as Surveyors, Navigators, Planners, Cartographers, or Reporters.

Audience:

Your classmates, teacher, and hypothetical community stakeholders who will review and evaluate your team's recommendations and solutions.

Situation:

Your barangay is planning projects that involve triangular spaces, such as garden plots, evacuation center roof trusses, and rice fields. Simultaneously, a rescue mission requires locating a missing vessel using bearings and distance calculations. Your team must apply trigonometric ratios, area formulas, and vector analysis to make accurate measurements, calculations, and practical recommendations.

Product/Performance:

A group presentation and report that includes:

1. Height measurements of a local landmark using trigonometric ratios.
2. Modeled triangular plots with calculated areas, estimated materials, and total costs.
3. Navigation map showing the rescue vessel's path, final position, and bearing.
4. Explanation of methods, reasoning, and real-life applications.
5. Reflection on teamwork and problem-solving strategies.

Standards:

- Correct use of trigonometric ratios to measure heights and calculate distances.
- Accurate calculation of triangle areas and estimation of materials and costs.
- Correct plotting of navigation paths, distances, and bearings.
- Clear, organized, and professional presentation of results.
- Active collaboration and effective communication among team members.
- Reflection demonstrates understanding of mathematical principles and practical applications.

Criteria	Weight	Beginning (1)	Developing (2)	Proficient (3)	Mastery (4)
1. Application of Trigonometric Ratios – Correctly uses trigonometric ratios to estimate heights or distances.	25%	Uses ratios incorrectly or not at all; calculations are mostly inaccurate.	Attempts to use ratios; some errors in setup or computation.	Correctly applies ratios with minor calculation errors.	Correctly applies ratios accurately and explains reasoning clearly.
2. Triangle Area Calculations & Material Estimates – Calculates triangle areas and estimates	25%	Miscalculates areas and materials; does not interpret results.	Partially correct calculations; minor errors in estimation or interpretation.	Mostly accurate area calculations and material estimates; interprets results appropriately.	Accurate calculations, clear estimation of materials/costs, and insightful interpretation for practical application.

Criteria	Weight	Beginning (1)	Developing (2)	Proficient (3)	Mastery (4)
materials and costs accurately.					
3. Navigation & Bearings – Correctly calculates distances, bearings, and plots positions for rescue mission.	20%	Miscalculates distances/bearings; path is inaccurate.	Partially correct calculations; plot is somewhat accurate.	Correct calculations with minor errors; plot mostly accurate.	Accurate calculations and bearings; final position and path plotted clearly and correctly.
4. Presentation & Communication – Clarity, organization, visual representation, and explanation of reasoning.	15%	The presentation is unclear, disorganized, and lacks visuals.	Somewhat clear; visuals are incomplete; reasoning partially explained.	Clear presentation with visuals; reasoning mostly explained.	Highly organized, clear, professional visuals; reasoning explained confidently and thoroughly.
5. Teamwork & Collaboration – Effective role distribution, participation, and reflection.	10%	Minimal participation; poor collaboration; reflection missing or vague.	Partial participation; uneven collaboration; reflection limited.	Good participation; collaborative; reflection shows understanding.	Excellent collaboration; all roles fulfilled; reflection insightful and demonstrates learning.
6. Reflection on Practical Application – Demonstrates understanding of how trigonometric concepts apply in real-life situations.	5%	Reflection missing or unrelated to activity.	Reflection shows limited understanding of application.	Reflection shows good understanding and connects to real-life contexts.	Reflection is thorough, insightful, and clearly connects mathematics to real-life applications.